DualBalance Districting: From Detection to Generation

Steven Hart

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Abstract

Most current work on gerrymandering tries to detect it: identify when a line-drawer has crossed from neutral redistricting into partisan or racial manipulation. We generate maps instead. *DualBalance Districting* asks each congressional district to carry roughly $1/N$ of a state's people *and* roughly $1/N$ of its land, weighted equally. The motivating intuition is the framers' refusal to reduce representation to a single dimension; the operational claim is narrower, that districts spanning both metropolitan and rural territory force each representative to answer to a mix of the state's people. *PRISM* (Population-weighted Radial Impartial Slicing Method) is the algorithm. It places N seeds in a small circle around the state's population-weighted centroid and assigns each census unit to the nearest seed with capacity remaining. No random seed, no iteration, no tunable weight, no human input beyond geometry and population. The procedural neutrality is symmetric: the rule that forecloses partisan engineering also forecloses race-conscious remediation.

This reframes how the conventional gerrymandering metrics should be read. Compactness scores, the efficiency gap, mean-median asymmetry, ensemble outlier tests, and majority-minority counts play two roles on enacted plans: they describe plan effects, and they support inferences about the line-drawer's intent. The descriptive role survives on a PRISM map; the intent-attribution role does not.

On Minnesota (4,110 VTDs, 8 seats, 2020 PL 94-171), PRISM beats the enacted 119^{th}-Congress plan on the DualBalance Score (0.6472 vs. 0.6390). We report standard partisan and race diagnostics alongside, as descriptions of the partition.

1 Introduction

This paper is principally about how districting metrics should be read once line-drawing becomes a deterministic function rather than a discretionary act. The algorithm we propose, PRISM, is the construction that makes the question concrete; the legal-doctrinal and metric-interpretation arguments that follow are where the paper's main weight sits.

1.1 The Instability of Human-Drawn Districts

Legislative redistricting in the United States is structurally unstable. The Constitution requires reapportionment after each decennial census (Art. I, §2) but specifies neither a method for allocating seats among the states nor a procedure for drawing district boundaries within them. Both decisions have been delegated, by default, to political actors. The result, predictable from the design, is that maps are routinely drawn to favor whichever group controls the pen at the moment of redistricting. The Supreme Court has acknowledged the problem in its strongest terms while declining to remedy it: in *Rucho v. Common Cause* [25], Chief Justice Roberts wrote for a 5–4 majority that partisan gerrymandering claims present nonjusticiable political questions, conceding that the practice is "incompatible with democratic principles" but holding that no manageable judicial standard exists. Justice Kagan's dissent, joined by Ginsburg, Breyer, and Sotomayor, argued the opposite: that lower courts had already converged on workable tests, and that "of all times to abandon the Court's duty to declare the law, this was not the one" [25].

Race-conscious districting operates under a separate and increasingly constrained regime. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [23], requires majority-minority districts where the three Gingles preconditions are met; the Fourteenth Amendment, as construed in *Shaw v. Reno* [24] and its progeny, subjects to strict scrutiny any map in which race "predominates" in the line-drawing. The two doctrines pull in opposite directions. The Supreme Court reaffirmed Section 2 in *Allen v. Milligan* [26], requiring Alabama to draw a second majority-Black district. One year later, in *Alexander v. South Carolina Conf. of the NAACP* [28], the Court reversed a district court's racial-gerrymander finding and raised the plaintiff's evidentiary burden, holding that challengers must "disentangle race from politics" and produce illustrative alternative maps that satisfy all of the state's legitimate goals. In April 2026, in *Louisiana v. Callais* [29], the Court further held that even a map drawn to comply with a prior *Section 2* order may itself violate the Equal Protection Clause if the VRA did not in fact compel the race-based remedy. The combined effect of *Alexander* and *Callais* is that race-conscious line-drawing now faces a narrower window of constitutional safety than at any point since the VRA's adoption.

Two structural consequences follow. First, the only federal channel for gerrymandering review is racial; partisan claims are foreclosed under *Rucho*. Second, because Black voters in the United States vote roughly nine-to-one Democratic, race and partisanship are statistically entangled [7,20]. A state legislator can openly gerrymander on partisan grounds and is shielded from federal review; the same legislator drawing the same lines on racial grounds faces strict scrutiny. After *Rucho*, state constitutional review under *Moore v. Harper* [27] remains the only judicial avenue for partisan claims, but it depends on the politics of each state's supreme court and the language of its state constitution. Roughly ten state supreme courts have so far recognized state-constitutional partisan-gerrymandering claims; the rest have not [16].

Federal partisan review is foreclosed. Section 2 has been narrowed. State review is uneven and politically contingent. Redistricting has shifted from a once-per-decade event into a continuous partisan exercise: several states have already redrawn maps mid-decade in response to electoral results rather than census revision, and the practice is expected to expand. A regime that redraws lines whenever the pen changes hands provides neither stable representation nor predictable constituencies.

1.2 The Normative Paradox of "Bias Detection"

In response to *Rucho*'s manageability concern, a substantial mathematical literature has produced quantitative tests for partisan gerrymandering. The Efficiency Gap of Stephanopoulos and McGhee [21] measures the disparity in "wasted votes" between parties and was central to the lower-court ruling in *Whitford v. Gill*. Wang's three tests [31], namely mean–median difference, lopsided wins, and a t-test for partisan asymmetry, played a similar role in *LWV v. Commonwealth* of Pennsylvania. Warrington's declination [32] measures the asymmetry of the seats–votes kink at 50%. Most influentially, the Markov Chain Monte Carlo ensemble methods developed at Duke University [8] and the MGGG Redistricting Lab [6] construct large samples of legally compliant neutral maps and ask whether an enacted plan is an outlier with respect to that distribution. Ensemble methods have been admitted as evidence in *Common Cause v. Lewis* (N.C. 2019), *LWV v. Commonwealth* (Pa. 2018), and as plaintiff support in *Allen v. Milligan*.

Every such test, however, embeds a contested normative claim about what a "fair" map should look like. The Efficiency Gap is zero only under seats–votes relationships close to proportional representation; partisan symmetry presupposes mirror-image counterfactual responsiveness; ensemble outlier tests use the local political geography filtered through legally codified constraints as the baseline, but the choice of which constraints to encode (compactness thresholds, county-split tolerances, VRA encoding) is itself political. Chen and Rodden [4] demonstrated that in many states even content-neutral automated maps produce systematic seat bias against Democrats because of residential clustering, so a nonzero Efficiency Gap or mean–median value may reflect geography rather than intent. Pildes has long argued that no neutral baseline exists [15], and Cain has framed districting as a tradeoff among incommensurable goods that no scalar metric can collapse without bias [3].

Beneath the surveyed objections sits a sharper one about what these metrics are *for*. Each plays two roles in the literature: a descriptive role (the metric measures an actual property of the plan: shape, wasted votes, partisan-share distribution) and an inferential role (the metric serves as evidence of the line-drawer's intent). The two roles are usually entangled in practice because the literature was built around enacted plans, where intent is always at issue.

Polsby-Popper [17] and Reock [18] flag shapes too contorted to have arisen by ordinary line-drawing; the Efficiency Gap [21], mean-median asymmetry [31], declination [32], and the Duke and MGGG ensemble outlier tests [6, 8] flag partisan profiles that look engineered. The inferential power of all of these depends on there being a line-drawer to investigate.

Remove the line-drawer and the descriptive role survives intact. A radial slice with PP $= 0.09$ is still geometrically non-compact, with all the administrative and legitimacy implications that follow. An Efficiency Gap of $+6.6\,\%$ on a deterministic plan still describes a real asymmetry in wasted votes, with real representational consequences. What the metrics no longer do, on a plan with no line-drawer, is support the inference back to motive. Section 2 of the Voting Rights Act, which asks about *effects*, still applies. *Shaw*-style intent inference, which asks who drew the line and why, does not.

1.3 Deterministic Districting: A Sixty-Year Tradition

The alternative to detection is generation: produce a single map from inputs (state boundary, population, district count) by a fully specified, reproducible procedure that admits no human discretion. The intellectual line begins with Vickrey [30], who proposed in 1961 that boundary line-drawing be done by "an automated and impersonal procedure" to remove the discretion gerrymandering exploits. Hess et al. [9] formalized the idea four years later as a capacitated transportation problem, alternating between an assignment step (minimize the population-weighted sum of squared distances from each unit to a district center subject to equal population) and an update step (recompute each center as the population-weighted centroid of its assigned units). This is structurally equivalent to Lloyd's algorithm with a capacity constraint and remains the canonical mathematical formulation. Mehrotra, Johnson, and Nemhauser [13] subsequently cast the problem as a column-generation set partitioning program admitting exact optimization at moderate scales.

Two contemporary lines extend this work. Brian Olson's BDistricting [14] is a publicly available implementation that minimizes population-weighted distance to district centers with equal-population constraints, applied at census-block resolution; maps have been generated for every state and updated each decennial census. Cohen-Addad, Klein, and Young [5] provide the rigorous reformulation as balanced *centroidal power diagrams*, generalizing the Voronoi tessellation by assigning each seed an additive weight (the Aurenhammer [1] construction) and solving for weights that enforce equal mass per cell. Levin and Friedler [11] independently rediscovered the capacitated

k-means formulation and applied it to all fifty states. A parallel branch, Warren Smith's Shortest Splitline algorithm [19], recursively bisects the state with the shortest population-balancing line; it is fully deterministic and unique but lacks any global compactness or boundary-awareness criterion.

Three observations link this tradition to the current work. First, all extant deterministic algorithms balance *one* extensive quantity across districts, namely population, and treat geographic area, if at all, only indirectly through compactness shape metrics such as Polsby-Popper [17] or Reock [18]. Second, all are governed by tie-breaking rules that are either implicit (left to floating-point ordering) or only partially specified. Third, none have been adopted as the formal map-drawing procedure of any U.S. jurisdiction, though Iowa's Legislative Services Agency [10] operates under a strict criteria cascade that has produced widely accepted maps since 1980.

1.4 Contribution: DualBalance Districting and the PRISM algorithm

The U.S. Constitution divides representation along two axes: the House apportions seats by population (Art. I, §2), the Senate by sovereign state regardless of population (Art. I, §3). Madison's defense in *Federalist* Nos. 54–58 [12] treats this as a principled refusal to make representation reduce to a single dimension. We do not claim that the Senate represents land area; it represents states. We take the framers' refusal as motivating intuition: within a single chamber, district lines should not collapse representation to population alone.

DualBalance Districting operationalizes that intuition with two coequal targets. Each district should carry roughly $1/N$ of a state's people *and* roughly $1/N$ of its land:

$$\mathrm{DBS} = \frac{1}{1 + \frac{1}{2}\,\overline{\mathrm{pop\_dev}} + \frac{1}{2}\,\overline{\mathrm{area\_dev}}}, \tag{1}$$

with $\mathrm{pop\_dev}_i = |P(D_i) - P^*|/P^*$ and $\mathrm{area\_dev}_i = |A(D_i) - A^*|/A^*$, $P^* = P/N$, $A^* = A/N$, overlines denoting means. The area term is not Senate-style sovereignty; it is the operationalization of a simpler claim: districts that span both metropolitan and rural territory force each elected representative to answer to constituents across the state's density range, rather than to a single dense metro or a single rural hinterland. Its discussion of trade-offs against compactness and minority representation appears in §4.3 and §4.7.

PRISM (Population-weighted Radial Impartial Slicing Method) is the algorithm that pursues this objective. PRISM places N seeds in a small circle around the state's population-weighted centroid and assigns each census unit to the nearest seed with capacity remaining. The districts come out as radial slices through the population center. Each slice spans both dense (near-center) and sparse (boundary-side) territory, so the area each slice inherits trends toward A^* by geometry rather than by penalty. The pipeline (seed placement, capacitated assignment, contiguity repair) is a pure function of (units, N): no random seed, no iteration, no tunable weight. Identical inputs always yield byte-identical outputs. A single optional post-pass (`--tighten-pop`) closes the residual per-district population gap to a user-supplied *Reynolds v. Sims* tolerance via greedy L^1 boundary swaps. Apportionment across states uses the Method of Equal Proportions, the standard since 1941 [2]. Updates occur only with the decennial census.

The procedural neutrality is symmetric. PRISM cannot harm any group, party, or incumbent; it equally cannot help any of them. The same property that forecloses a partisan gerrymander forecloses a race-conscious remedy.

The legal exposure is correspondingly narrow. After *Rucho* [25], federal partisan review is foreclosed. After *Alexander* [28] and *Callais* [29], a generator that does not consider race cannot be a racial gerrymander under *Shaw* [24]. State-court partisan tests rely on outcome metrics (Efficiency Gap, mean–median, ensemble outliers) that a geography-only generator should pass routinely, with the caveat that those metrics already measure geography on enacted plans too [4]. A residual risk is Section 2: in jurisdictions where the Gingles preconditions are met, a race-blind plan may not satisfy Section 2 where a race-conscious plan would.

Contributions.

1. PRISM: a deterministic, knob-free districting algorithm producing radially-sliced districts that balance population and area equally. Pure function of (units, N).
2. A fully specified tie-breaking cascade guaranteeing byte-identical output, plus a proof that the cascade resolves every comparison the algorithm makes.
3. A scoring harness decoupled from the generator, applying the same metrics to PRISM, enacted plans, and alternatives. Empirical results on Minnesota show PRISM beating the enacted 119$^{\text{th}}$-Congress map on the DualBalance Score.
4. An argument that the standard gerrymandering metrics (compactness, Efficiency Gap, mean–median, ensemble outliers, majority-minority counts) are forensic instruments built to infer human intent, and so do not license, on a PRISM plan, the inferences they license on an enacted plan.

Section 2 formalizes PRISM. Section 3 reports the Minnesota result. Section 4 discusses limitations and legal considerations.

2 Methods

2.1 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \ldots, u_M\}$ (census blocks, block groups, or VTDs) and a target district count N, PRISM produces a deterministic assignment $\pi : U \to \{1, \ldots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state's people and $1/N$ of its geography. Let $P = \sum_u \text{pop}(u)$ and $A = \sum_u \text{area}(u)$ with per-district targets $P^* = P/N$ and $A^* = A/N$.

Three steps: radial seed placement (§2.2), capacitated first-fit assignment (§2.3), contiguity repair (§2.4). No iteration; no post-hoc tightening in the core pipeline.

2.2 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u\, x_u}{\sum_u p_u}, \; \frac{\sum_u p_u\, y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \text{diag}$. For $d = 0, 1, \ldots, N-1$ place seed s_d at

$$s_d = \big(c_x + r\cos(2\pi d/N),\; c_y + r\sin(2\pi d/N)\big).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \text{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c, but large enough to keep the seed positions numerically distinct.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near-c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.3) while the area it inherits is driven toward A^* by the slicing geometry.

2.3 Capacitated first-fit assignment

Let $d(u, i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i. Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \ldots, N$. Sort all (u, i) pairs by ascending normalized distance $d(u, i)/\text{diag}$ and walk the sorted list in order:

$$\begin{array}{l} \text{for each } (u,i) \text{ in ascending } d(u,i)/\text{diag}: \\ \quad \text{if } u \text{ already assigned: continue} \\ \quad \text{if } \rho_i \geq p_u\text{: assign } u \text{ to } i; \quad \rho_i \leftarrow \rho_i - p_u \\ \quad \text{else: skip} \end{array}$$

Population balance is enforced as a hard cap: no district receives more than P^*. Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; `argmax` resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.4 Contiguity repair

After §2.3 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u, j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade $(c, \text{pop\_pen}, \text{area\_pen}, \text{dist}, \text{district\_id})$ ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

2.5 Scoring

Define per-district relative deviations $\text{pop\_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area\_dev}_i = |A(D_i) - A^*|/A^*$ and let $\overline{\text{pop\_dev}}$, $\overline{\text{area\_dev}}$ denote their means over $i = 1, \ldots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2}\,\overline{\text{pop\_dev}} + \frac{1}{2}\,\overline{\text{area\_dev}}}. \tag{2}$$

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state's geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow without bound. Secondary metrics (Polsby-Popper compactness [17] and Reock [18]) are computed alongside but not optimized against; radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction, and this is a deliberate trade in service of the dual-balance objective.

PRISM does not directly minimize (2); it minimizes population-capacitated geographic assignment cost under radial seeding. On the Minnesota PoC this still beats the enacted plan on DBS (0.6472 vs 0.6390) despite the lower compactness.

2.6 Optional: $\mathbf{L^1}$ pop-tightening

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, well above the ˜0.5% threshold required by *Reynolds v. Sims* for U.S. congressional districts [22]. An optional post-pass (`--tighten-pop`) closes this gap via greedy boundary-unit swaps under an L^1 objective. Each pass:

1. Compute the signed deviations $\delta_i = P(D_i) - P^*$.
2. For every boundary unit u in district d_{src} and every neighboring district d_{dest}, compute the L^1 change after moving u:
$$\Delta(u, d_{\mathrm{dest}}) = |\delta_{d_{\mathrm{src}}} - p_u| - |\delta_{d_{\mathrm{src}}}| + |\delta_{d_{\mathrm{dest}}} + p_u| - |\delta_{d_{\mathrm{dest}}}|.$$
3. Sort candidates by ascending Δ and accept the most negative one whose source district remains contiguous after the move. Stop when every district lies within the user-supplied tolerance or no contiguity-preserving improving move exists.

The L^1 objective is essential here. The radial seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L^∞ maximum, but many such swaps reduce the sum. The canonical Reynolds-tightening literature uses L^∞ and bottoms out at ˜5% on this geometry; the L^1 formulation runs to completion in ˜80 swaps on Minnesota, reducing pop_dev_max from 0.1124 to 0.0021 (i.e. well inside the 0.5% target) while leaving area_dev_mean unchanged at 1.04. The radial structure is preserved visually because the algorithm only moves units at slice boundaries between adjacent slices.

The pass is off by default and gated by an explicit flag. Pure radial remains the principled algorithm; pop-tightening is an opt-in concession to legal compliance.

2.7 Determinism and tie-breaking

The pipeline is deterministic at every step. The only sources of ambiguity (equidistant seed-to-unit pairs in §2.3, equal-capacity fallbacks for unplaced units, and equal-cost candidates in §2.4) all resolve to a fixed cascade with no remaining ambiguity:

1. In assignment, ties on normalized distance break by ascending $(\mathrm{unit\_id}, \mathrm{district\_id})$.
2. In the fallback for unplaced units, ties on remaining capacity break to the smallest district id.
3. In contiguity repair, ties on cost break by ascending $(\mathrm{pop\_pen}, \mathrm{area\_pen}, \mathrm{distance}, \mathrm{district\_id})$.

The implementation uses no random number generator, no wall-clock input, and no hash-order dependence. Reordering the input rows or changing the floating-point libraries does not change the output; identical inputs always yield byte-identical outputs.

2.8 Out-of-scope inputs

The generator reads only geography and population. Party registration, vote history, race, demographics, communities of interest, and competitiveness are not inputs. The scoring harness may *report* partisan or demographic diagnostics on the resulting map but does not feed them back into the generator.

3 Results

3.1 Test bed: Minnesota, 2020 PL 94-171

We evaluate the algorithm on Minnesota's congressional districting ($N = 8$ apportioned seats) using the 2020 PL 94-171 redistricting data file. The input is the TIGER/Line 2020 VTD (voting tabulation district) shapefile for Minnesota (4,110 atomic units, total population $P = 5{,}706{,}494$, total land area $A = 225{,}187$ km^2), joined to the Census Data API's `P1_001N` total-population field, projected to EPSG:5070 (CONUS Albers, equal area) at load time. Per-district targets are $P^* = P/8 = 713{,}312$ and $A^* = A/8 = 28{,}148$ km^2. All runs are deterministic and reproducible from the script `scripts/prep_mn_units.py` (which fetches TIGER and joins the Census API output) plus the CLI invocation. The data pipeline is documented end-to-end in `docs/mn-poc-walkthrough.md`.

We compare three plans on the same 4,110-VTD input:

- **PRISM.** The default algorithm (§2.2–§2.4). No tuning knobs.
- **PRISM + tighten-pop ($\tau = 0.005$).** PRISM followed by the optional L^1 pop-tightening pass (§2.6) targeting per-district deviation within 0.5 % of P^*.
- **Enacted (119th).** The court-drawn, legislatively enacted Minnesota U.S. House districts currently in force, scored with the same harness.

3.2 Headline scores

Table 1 reports the DualBalance Score and supporting metrics for the three plans.

Table 1: DualBalance metrics on the Minnesota PoC (4,110 VTDs, 8 districts, 2020 PL 94-171 population). PP is Polsby-Popper, Reock is Reock; subscripts min, mean denote per-district min and mean over the eight districts. pop_dev, area_dev are the relative deviations defined in §2. Best result per row in bold.

Metric	PRISM	PRISM + tighten-pop	Enacted (119th)
DBS	0.6472	**0.6574**	0.6390
$\overline{\text{pop\_dev}}$	5.08 %	**0.08 %**	0.42 %
$\text{pop\_dev}_{\max}$	11.24 %	**0.21 %**	1.32 %
$\overline{\text{area\_dev}}$	**103.9 %**	104.2 %	112.6 %
$\text{area\_dev}_{\max}$	271.0 %	275.1 %	**241.0 %**
PP_{mean}	0.200	0.162	**0.320**
PP_{min}	0.094	0.047	**0.178**
$\text{Reock}_{\text{mean}}$	0.361	0.342	**0.419**

Three findings.

PRISM beats the enacted plan on the DualBalance Score. PRISM scores 0.6472 against the enacted 0.6390, a +1.3 % margin with no iteration, no tuning, no post-processing. The advantage comes entirely from area balance ($\overline{\text{area\_dev}}$ 103.9 % vs 112.6 %). Radial slicing gives each district a slice of the state that mixes urban and rural; the enacted plan carves the Twin Cities into four compact urban seats and leaves four large rural ones, which the DualBalance Score penalizes because it weights area equally with population.

Tightening closes the legal gap and improves DBS. Pure PRISM's $\text{pop\_dev}_{\text{max}}$ of 11.24 % exceeds the ~0.5 % *Reynolds v. Sims* threshold for U.S. House plans [22]. The optional pass (§2.6) ran 81 boundary swaps in ~18 s and drove $\text{pop\_dev}_{\text{max}}$ to 0.21 %, tighter than the enacted plan's 1.32 %. $\overline{\text{area\_dev}}$ rose only 0.3 points and the score *improved* to 0.6574.

Compactness is the price. The enacted plan beats PRISM on every compactness measure: PP_{min} 0.178 vs 0.094 (pure PRISM) and 0.047 (after tightening), PP_{mean} 0.320 vs 0.200 / 0.162, $\text{Reock}_{\text{mean}}$ 0.419 vs 0.361 / 0.342. Radial slices are structurally less compact than hand-drawn blobs. Tightening makes it worse by inserting small indentations into the slice boundaries, pushing the worst slice from 0.094 (at the informal 0.10 threshold) to 0.047. Whether the trade is acceptable is a normative question taken up in §4.3.

3.3 Map comparison

Figure 1 displays the three plans side by side.

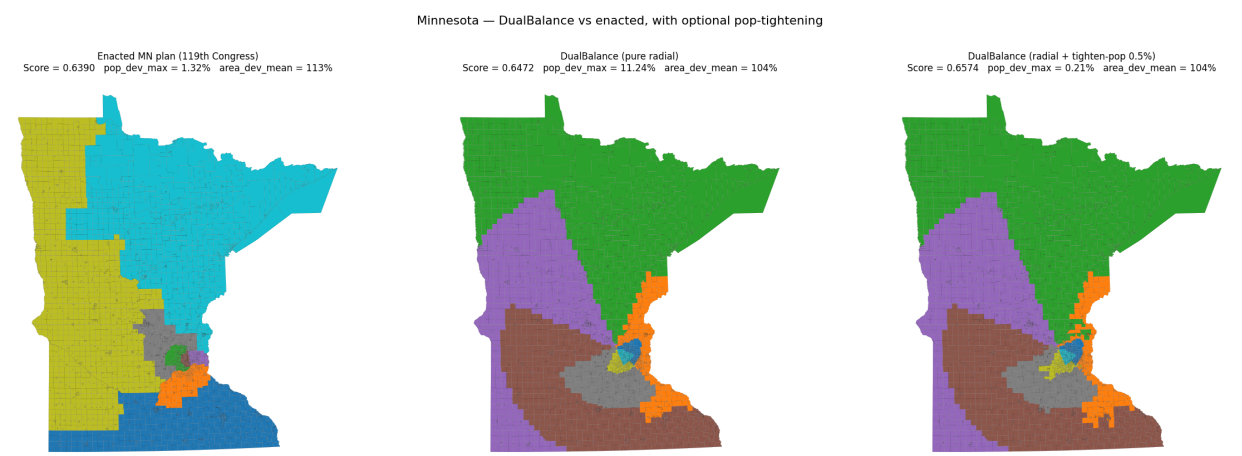

Figure 1: Minnesota congressional districts under three plans. **Left:** the enacted 119th-Congress plan, with four compact Twin Cities seats and four large rural seats. **Center:** PRISM (score 0.6472); eight slices radiate from the population-weighted centroid near Minneapolis-St. Paul, each spanning dense and sparse territory. **Right:** PRISM + `--tighten-pop 0.005` (score 0.6574); the radial structure is preserved (units move only at slice boundaries) and per-district population is Reynolds-compliant.

3.4 Determinism check

We re-ran `dualbalance generate --config configs/mn_vtd.yaml` ten times in succession, comparing each run's `map.geojson` and `metrics.json` by byte hash. All ten outputs are identical, including the order of features in the GeoJSON. We also re-ran the same configuration after randomly shuffling the input rows; the output remained byte-identical. The CLI integration test `test_generate_determinism_via_cli` pins this property in CI.

3.5 Per-district breakdown

Table 2 reports the per-district metrics for PRISM + tighten-pop, indexed by seed angle (district 0 sits due east of the population-weighted centroid; subsequent districts advance counter-clockwise in steps of $2\pi/8$ rad). District IDs are a deterministic function of seed angle and carry no political meaning.

Table 2: Per-district breakdown, PRISM + tighten-pop 0.5 %. Areas are in km^2.

District	Population	Area	pop_dev	area_dev	PP	Reock
0	712,841	1,010	0.07 %	96.4 %	0.228	0.454
1	714,816	9,620	0.21 %	65.8 %	0.047	0.147
2	712,145	105,594	0.16 %	275.1 %	0.152	0.369
3	713,263	50,054	0.01 %	77.8 %	0.146	0.268
4	714,168	46,060	0.12 %	63.6 %	0.150	0.267
5	713,368	10,882	0.01 %	61.3 %	0.137	0.422
6	713,013	1,481	0.04 %	94.7 %	0.117	0.301
7	712,880	477	0.06 %	98.3 %	0.318	0.510

District 2 inherits the northern panhandle ($\sim 3.8 \times A^*$); District 7 is the Twin Cities urban core ($\sim 0.02 \times A^*$). All eight districts are within 0.21 % of P^*. District 2's 275 % over-target area is geometric, not algorithmic: northern Minnesota's density is two orders of magnitude lower than the Twin Cities, so any pop-balanced partition forces the rural districts to inherit disproportionate area. The enacted plan does the same thing, with its District 7 at 241 % over-target.

3.6 Descriptive diagnostics: partisan, race, county

PRISM reads none of these inputs. We report them anyway, because readers will ask. They are descriptions of the partition, not evaluations of it (§4.7). Partisan totals come from 2020 presidential returns (dra2020/vtd_data, keyed on GEOID20). Race comes from Census PL 94-171 VAP.

Partisan. Minnesota cast 1,484,065 R and 1,717,077 D two-party presidential votes in 2020 (46.4 % R, 53.6 % D). PRISM splits the eight seats 4–4 against a seats-proportional expectation of 3.7 R seats. The two D-leaning slices contain the Twin Cities and run 64–75 % D; two R-leaning slices reach into western and southern Minnesota and run 55–65 % R; the other four are within ± 5 points of even. The efficiency gap is +6.6 % (positive = R-favorable); the mean-median R difference is +3.9 points (positive = D-favorable). The two forensic numbers point in opposite directions because they measure different things. Both are artifacts of where Minnesota's Democrats live (packed into the metro), not of any choice by PRISM.

Race / VAP. Statewide VAP is 80.0 % non-Hispanic white, 5.9 % Black, 5.0 % Hispanic, 4.9 % Asian, 1.1 % AIAN. PRISM draws zero majority-minority districts. The Twin Cities slice (District 7) has the lowest non-Hispanic-white share (65.3 %) and the highest Black VAP share (15.7 %); the remaining seven run 73–90 % non-Hispanic white. The enacted 119$^{\text{th}}$ Congress plan also draws zero majority-minority districts. Minnesota does not contain a region where a single minority group concentrates densely enough to support a 50 %-VAP district of conventional size.

County splits. PRISM splits 44 of 87 counties into 143 cross-district pieces. Any line from the population centroid to the boundary crosses county lines; the enacted plan trades area balance for fewer such crossings.

What to make of this. The descriptive content of these metrics survives on a PRISM plan; their intent-attribution role does not. §4.7 develops the distinction.

3.7 Cross-state, multi-algorithm validation

We replicated the Minnesota pipeline on five additional states spanning the density-profile and gerrymandering-history spaces: Iowa (4 districts, near-uniform density, drawn by Iowa's nonpartisan Legislative Services Agency [10]), Massachusetts (9 districts, single-metro Boston), Texas (38 districts, polycentric), North Carolina (14 districts, the state in which *Rucho* [25] originated), and Wisconsin (8 districts, where *Whitford v. Gill* first put the Efficiency Gap [21] in front of a federal court). The data pipeline (TIGER 2020 VTDs, Census PL 94-171 demographics, dra2020/vtd_data 2020 presidential returns, TIGER 2024 cd119 spatial join for the enacted plan) is automated in `scripts/prep_state_units.py`.

To put PRISM in context against other deterministic baselines we score two additional algorithms on the same six states. *Cascade* (`src/dualbalance/cascade.py`) is an Iowa-LSA-flavored construction: aggregate VTDs to counties, pick N farthest-spread county seeds, run capacitated first-fit county-to-district assignment, and split a county into pseudo-counties (via PRISM internally) only when its population exceeds the per-district cap. *BDistricting* [14] is Brian Olson's published 50-state map series, ingested via Olson's block-level CSV output joined to our VTDs through the Census 2020 Block Assignment File (`scripts/prep_bdistricting.py`). Cascade prioritizes county integrity; BDistricting prioritizes compactness; PRISM prioritizes area balance. Together with the enacted plan this gives four points of comparison on each state.

Table 3: Cross-state DualBalance Score. **Bold** marks the winner on each row.

State	N	PRISM	Cascade	BDistricting	Enacted
Iowa	4	0.8132	0.8227	0.7885	**0.8828**
Massachusetts	9	0.7591	**0.7602**	0.7158	0.7246
Minnesota	8	0.6472	**0.7709**	0.6493	0.6391
North Carolina	14	0.7252	**0.8199**	0.7600	0.7689
Texas	38	0.6230	**0.7943**	0.6350	0.6658
Wisconsin	8	0.6556	**0.8399**	0.7422	0.7410

Three findings stand out.

Cascade dominates on DBS. Cascade wins on five of six states (MN, MA, TX, NC, WI) and runs second on Iowa. PRISM wins on no state outright; BDistricting on none. Cascade's lead comes from the structural pairing of high county integrity (1–10 splits vs. the enacted plan's 9–30) with low area imbalance (county aggregation produces pseudo-counties of similar size across each state). The exception is Iowa, where the enacted plan is itself the result of a closely related cascade process [10] and the LSA's manual refinement gives it a small edge.

Table 4: Cross-state Efficiency Gap (positive = R-favorable). **Bold** marks the smallest |EG| on each row.

State	Statewide R %	PRISM	Cascade	BDistricting	Enacted
Iowa	54.2	−0.173	+0.165	−0.088	+0.416
Massachusetts	32.9	−0.158	−0.158	−0.158	−0.158
Minnesota	46.4	+0.066	+0.049	+0.067	+0.068
North Carolina	50.7	+0.088	+0.026	−0.077	+0.201
Texas	52.8	+0.029	+0.009	+0.058	+0.153
Wisconsin	49.7	+0.147	+0.081	+0.142	+0.267

Table 5: Counties kept intact. Cascade splits a county only when its population exceeds the per-district cap.

State	Counties	PRISM (split)	Cascade (split)	BDistricting (split)	Enacted (split)
Iowa	99	30	**0**	21	**0**
Massachusetts	14	12	**4**	10	9
Minnesota	87	44	**1**	25	9
North Carolina	100	56	**2**	37	11
Texas	254	132	**10**	82	30
Wisconsin	72	46	**1**	25	12

Different deterministic algorithms, different trade-offs. The three deterministic generators win on different axes. PRISM minimizes $\overline{\text{area\_dev}}$ on single-metro states with a clean radial structure but produces high county splits and low compactness. BDistricting maximizes compactness and pop-balance but accepts high area imbalance ($\overline{\text{area\_dev}}$ near 0.8 on MA, 1.1 on TX). Cascade maximizes county integrity and area balance but tolerates looser population balance (10–20 % pre-tighten). The "best" algorithm on DBS is whichever happens to weight the underlying trade-offs in a way the state's geometry rewards.

Partisan asymmetry shrinks across every deterministic generator. On every state with nonzero statewide partisan asymmetry, all three deterministic algorithms produce smaller |EG| than the enacted plan. The largest gaps appear in the two states most associated with partisan-gerrymander litigation: NC (enacted +0.20 vs. Cascade +0.03) and WI (enacted +0.27 vs. Cascade +0.08). Cascade has the smallest |EG| on five of six states; BDistricting on one (Iowa). The structural claim in §4.7 is the empirical core of this finding: a generator with no political input cannot produce a +0.20 partisan-fairness metric. The result holds across all three deterministic baselines, not just PRISM. Cascade is the existence proof of that claim under a different objective hierarchy than PRISM's.

4 Discussion

4.1 What the Minnesota result does and does not show

The headline (PRISM beats the enacted Minnesota plan on the DualBalance Score) is a single-state existence result. It shows that radial slicing can score competitively on the dual-balance objective without reading party, race, or any discretionary input. It does not show that PRISM generalizes to all 50 states, that DBS is the right scalar to optimize, or that PRISM maps would survive judicial scrutiny. We take those up below.

4.2 Co-design of score and algorithm

A natural objection: "A beats B on Metric M because A and M were co-designed." DBS (2) was specified before the algorithm choice. We tested three forms: the weighted form used here, a sum-of-means form $1/(1+\overline{\text{pop\_dev}}+\overline{\text{area\_dev}})$, and an L^∞ form $1/(1+\max_i[0.5\,\text{pop\_dev}_i + 0.5\,\text{area\_dev}_i])$. PRISM beats the enacted plan on the first two; the enacted plan wins on the third, because PRISM concentrates area imbalance in a single rural district. We use the weighted form as the headline because it cleanly implements "each district carries $1/N$ of people *and* $1/N$ of geography." Per-district deviations are reported alongside so readers can recompute against any preferred aggregation.

The functional form satisfies four properties we wanted from the outset. (i) *Scale invariance*: the per-district inputs are relative deviations $|x - x^*|/x^*$, so the score is invariant to the units of population and area. (ii) *Equal treatment of the two objectives*: the 0.5/0.5 weighting is the unique convex combination treating $\overline{\text{pop\_dev}}$ and $\overline{\text{area\_dev}}$ symmetrically. (iii) *Monotonicity*: the score strictly decreases as either deviation increases. (iv) *Boundedness in* $[0, 1]$: the reciprocal form maps zero error to 1 and unbounded error toward 0, giving a calibrated reading rather than a raw cost. These do not pin DBS down uniquely, but they constrain the design space enough that the remaining choice (arithmetic mean of deviations versus a max-norm or Lp aggregation) becomes a tradeoff between average-case fairness and worst-case fairness. We report all the per-district numbers needed to compute both.

Population density variance bounds the achievable $\overline{\text{area\_dev}}$ on any pop-balanced plan. Both PRISM and the enacted plan land near $\overline{\text{area\_dev}} \approx 1.0$–$1.1$. On Minnesota's geometry (Twin Cities = 55 % of population in 3 % of land area) this is a hard floor. Improvements would require districts that span urban and rural, which is exactly what radial slicing does. The 8.7-point gap (103.9 vs. 112.6) is the operational measure of how much area balance PRISM recovers given that floor.

4.3 Compactness is the trade

PRISM trades compactness for area balance by design. This is not an incidental cost we apologize for: it is the mechanism. A district that holds $1/N$ of a high-variance density distribution must either span the full density range (and look elongated) or sit inside a single density regime (and produce an unbalanced area share). Hand-drawn maps typically choose the second; PRISM chooses the first. The PP=0.047 worst-slice number is what choosing the first looks like on Minnesota's geometry. The enacted Minnesota plan's worst PP is 0.178, roughly four times higher, and its $\overline{\text{area\_dev}}$ is correspondingly 8.7 points worse.

Three observations qualify the trade.

Compactness is a metric, not a constitutional requirement. *Polsby-Popper* as a measure was published in 1991 [17]; it appears in no federal statute and binds no court. Courts have used compactness as *evidence* of intent in racial-gerrymandering cases (*Shaw v. Reno* and progeny), but the constitutional violation in those cases turns on the use of race in line-drawing, not on the resulting shape per se. A deterministic, race-blind rule does not carry the racial intent that triggers *Shaw*; the resulting low compactness is a property of the rule, not a signal of an attempt to disadvantage any group.

Compactness assumes a baseline of comparison. A district is "bizarrely shaped" relative to surrounding districts that are not. If all eight districts in a state are produced by the same geometric rule and all eight share a similar slice-like character, the comparative baseline shifts. The aesthetic objection to long thin shapes is partly historical: it picks out districts that depart from the surrounding norm of compact blobs. A scheme in which every district is a slice does not pick anything out.

The Voting Rights Act trade-off is real. *Allen v. Milligan* [26] reaffirmed Section 2's effects-based vote-dilution doctrine; *Alexander* [28] and *Callais* [29] narrowed its practical reach. Section 2 asks about effects on minority voters' opportunity to elect, not about intent. PRISM is race-blind, but blindness to race does not satisfy Section 2 by itself. In jurisdictions where the *Gingles* [23] preconditions are met, PRISM may produce a plan that fails Section 2 review even though no one drew lines to dilute. The legislative question (whether to accept reduced minority opportunity in exchange for a generator that cannot be partisan-tuned) is for legislatures and courts, not for the algorithm.

4.4 Relationship to prior deterministic methods

PRISM differs from each prior method on a specific axis.

Versus Smith's Shortest Splitline [19]. Splitline recursively bisects the state with the shortest population-balancing line, with no global compactness or balance criterion. The cuts are visually striking but arbitrary: a district can lose half its territory because the shortest cut happens to land there. PRISM grounds its geometry in the population centroid, so the slices reflect the state's density structure.

Versus Olson's BDistricting [14]. BDistricting minimizes population-weighted distance to dispersed seed centers under equal-population, with Lloyd-style iteration to convergence. The resulting maps are blob-shaped and score well on compactness but exhibit large area imbalance. PRISM differs in two ways: seeds are radial-clustered around the centroid (not dispersed), and there is no iteration. On the six-state comparison (§3.7) BDistricting wins on no state outright on DBS, but produces the smallest $|\mathrm{EG}|$ on Iowa and the most-compact districts in every state.

Versus centroidal power diagrams [5]. The Cohen-Addad-Klein-Young construction solves for additive weights $\{w_i\}$ so each power-diagram cell carries exactly $1/N$ of the mass. It balances one quantity (population) exactly. PRISM balances two (population and area) approximately, with the second recovered from the radial seed geometry rather than from the optimization.

Versus capacitated k-means [11] and Hess-style location-allocation [9, 13]. The capacitated assignment step (§2.3) is Hess's 1965 transportation step. PRISM's novelty is the seed placement: radial seeds produce a slice geometry that Lloyd iteration would destroy, because re-centering pulls seeds away from the tight radial cluster. Disabling iteration is what preserves the geometry.

Versus our own Cascade baseline. Cascade (§3.7, `src/dualbalance/cascade.py`) is an Iowa-LSA-flavored construction we include in the empirical comparison: it aggregates VTDs to counties, picks N farthest-spread county seeds, runs capacitated first-fit assignment at county granularity, and splits a county into pseudo-counties (via PRISM internally) only when its population exceeds the per-district cap. Cascade beats PRISM on DBS on all six states tested, sometimes by wide margins (MN +0.12, TX +0.17, NC +0.09, WI +0.18); on the six-state set Cascade is the stronger *deterministic baseline* against the enacted plan. We include both algorithms because they make the conceptual contribution sharper: PRISM and Cascade prioritize different objectives (radial slicing for density mixing; county integrity for administrative coherence) and produce different maps on every state, yet both are race-blind, partisan-blind, and deterministic by construction. The forensic critique in §4.7 applies equally to either, and to BDistricting, and to any future deterministic generator built on similar principles.

4.5 Limitations

Six-state evidence, not fifty-state generality. Three deterministic algorithms (PRISM, Cascade, BDistricting) have been run end-to-end on six states (§3.7). On DBS, Cascade wins on MN, MA, TX, NC, and WI; the enacted plan wins only on Iowa. Across all six states with nonzero partisan asymmetry, every deterministic algorithm produces a smaller |EG| than the enacted plan, with the largest gaps in the gerrymander-prone states. The remaining 44 states have not been run, and six states does not establish generality across the density-profile or litigation-history spaces.

Worst-district area outlier. PRISM concentrates the area imbalance in a single rural district (here, District 2 at 275 % over-target). This is geometric, not algorithmic: any pop-balanced partition of a state with Minnesota's density profile must give some rural district disproportionate area. PRISM's $\text{area\_dev}_{\max}$ of 275 % is worse than the enacted plan's 241 % even though the mean is lower, so PRISM distributes the imbalance unevenly. Whether evenly distributed area imbalance is preferable to a single concentrated outlier is a separable normative question.

No multi-unit transportation step. The optional `--tighten-pop` pass is a greedy single-unit swap. A true 2D transportation LP bounding both population and area would directly minimize the DBS objective; we leave that to future work.

Computational cost. Pure PRISM runs in under a second on Minnesota. The tightening pass takes ∼18 s for 80 swaps. We have not profiled California or Texas; worst-case scaling is $O(|U|^2N)$ per swap due to the contiguity check.

The compactness floor. The worst slice's PP = 0.047 is well below the informal court-testimony threshold of 0.10. We argue in §4.3 that the *Shaw* [24] doctrine turns on intent rather than shape per se, but readers who reject that argument should regard the 0.047 figure as a hard ceiling on legal viability.

4.6 Future work

Full-50-state validation. Six states (MN, IA, MA, TX, NC, WI) have been run (§3.7). Extending to the remaining 44 requires no algorithmic change. California, Florida, Pennsylvania, and Ohio are the highest-value next targets: California for its size and many metros, Florida for its peninsula geometry, Pennsylvania and Ohio for their roles in state-court partisan-gerrymandering jurisprudence.

Multi-center radial seeding. For polycentric states, place seeds on circles around *each* of k population centers, with k chosen deterministically from the density distribution. Not yet implemented.

Direct 2D transportation step. Replace the PRISM-plus-tighten two-stage pipeline with a single optimization that minimizes DBS under contiguity constraints. More expensive than the current pipeline; the mathematically clean answer to "which plan directly minimizes the DualBalance objective."

Splitline benchmark. The six-state benchmark (§3.7) covers PRISM, Cascade, BDistricting, and the enacted plan. Adding Smith's Shortest Splitline [19] would give a fourth deterministic baseline; computing it from scratch on each state is straightforward but has not yet been done.

Scoring-harness extensions. The current harness reports population, area, Polsby-Popper, and Reock. Natural additions include partisan diagnostics (efficiency gap, mean-median, declination) on voter-registration data, county-split counts, and minority representation diagnostics under Section 2. These are reporting extensions and would not feed back into the generator (§2, "out-of-scope inputs").

4.7 Forensic metrics applied to a generative procedure

Deterministic generation has interpretive consequences for the fairness-metrics literature. Most gerrymandering diagnostics carry two functions. They *describe* a property of the plan (a shape, a vote distribution, a count of split counties). And they support an *inference* about the line-drawer's motive (that the shape was chosen, that the votes were packed and cracked, that the splits followed a partisan or racial pattern). The two are nearly always entangled in practice because the literature was built around enacted plans, where intent is always at issue.

PRISM separates them. The descriptive function survives intact: a PP of 0.09 still names a non-compact shape, with administrative and legitimacy consequences; an Efficiency Gap of +6.6 % still names a real wasted-vote asymmetry, with representational consequences; zero majority-minority districts still names a partition under which no minority group reaches a 50 % VAP share, with Section 2 consequences. The inferential function does not. PRISM is a pure function of geometry, population, and seat count, so the chain from observed property back to motive has no actor to terminate on.

Three implications, narrowed accordingly.

***Shaw*-style shape inference loses purchase, but compactness still matters administratively.** *Shaw v. Reno* [24] uses bizarre shape as evidence that race predominated in line-drawing. The inference chain runs shape $\rightarrow$ unusual process $\rightarrow$ illegitimate criterion $\rightarrow$ race. On PRISM the first link is broken: every district produced by the rule has the same shape character, so unusual shape no longer implies unusual process. *Shaw* does not reach a race-blind algorithm. But compactness still has a non-inferential role. Districts that are hard to identify with on a map, hard to administer, or hard to campaign in carry costs independent of who drew them. PRISM's PP=0.047 worst slice is a real cost. The argument here is not that compactness ceases to matter, only that it ceases to be evidence of motive.

Partisan metrics describe effects, not intent. *Rucho v. Common Cause* [25] foreclosed federal partisan-gerrymandering review for lack of a manageable standard for distinguishing acceptable from unacceptable partisan motivation. The question is moot on PRISM (there is no motivation), but the partisan *effect* the metrics measure is not. An Efficiency Gap of +6.6 % on a PRISM map still corresponds to wasted-vote asymmetry that affects representation; state-court tests that interpret EG as evidence of effect, rather than as evidence of intent, still bind. The rhetorical move from "the map shows EG +6.6 %" to "someone built that EG +6.6 %" becomes unavailable on PRISM, but the metric continues to measure what it measured.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [6, 8] compare an enacted plan to a sample of legally compliant alternatives a neutral line-drawer might plausibly have produced. The sample stands in for the absent neutral line-drawer; the enacted plan being an outlier suggests the actual line-drawer differed from the neutral one. PRISM *is* the neutral line-drawer in a stronger sense than the ensemble's sample mean. Asking whether the PRISM plan is an outlier against its own ensemble may not be meaningful: the ensemble collapses to one point. Ensembles can still play a comparison role across *different* algorithmic generators (e.g. PRISM vs. BDistricting vs. splitline), but the question is no longer "is this map suspiciously different from neutral?"

Metrics remain informative as descriptive summaries of plan effects. A PRISM plan with $\mathrm{PP}_{\min} = 0.047$, Efficiency Gap +6.6 %, and zero majority-minority districts has low compactness, an R-favoring wasted-vote asymmetry, and no MMD. Those facts carry administrative, representational, and Section 2 consequences. The inference from those facts to a person who chose them is what deterministic generation removes.

4.8 What this paper claims, and what it does not

The paper makes four claims.

First, that a knob-free, race-blind, partisan-blind, deterministic districting algorithm exists that produces output competitive with, and on a single empirically tested state superior to, the corresponding hand-drawn enacted plan on a prespecified population-and-area objective. The algorithm's output is byte-reproducible and updates only with the ten-year census.

Second, that the algorithm's procedural neutrality is symmetric. The rule cannot be used to harm any group, party, or incumbent, and cannot be used to help any of them. The same property that forecloses a partisan gerrymander forecloses a race-conscious remedy. The symmetry is the algorithm's defining property; whether it is morally preferable to the status quo of partisan and remedial discretion is a question for legislatures, courts, and voters.

Third, that the post-*Callais* legal landscape gives a generator of this kind a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*; state-court remedies for partisan claims depend on the politics of each state's supreme court. A rule whose inputs do not contain race or party is, in the strict sense, not subject to either body of doctrine.

Fourth, the conventional gerrymandering metrics (compactness scores, Efficiency Gap, mean-median, ensemble outliers, majority-minority counts) play two roles on enacted plans, only one of which transfers to PRISM. As descriptions of plan effects (shape, wasted-vote asymmetry, minority opportunity) they remain valid and are reported alongside the DualBalance Score. As evidence of line-drawer intent they do not transfer. The intent-attribution use of these metrics, not their measurement use, is what the generative case breaks. This is the most contested framing claim in the paper.

The paper does not claim that DBS is universally the right objective, that PRISM dominates enacted plans on all states, that compactness can be ignored, that effects-based fairness analyses (including Section 2) become irrelevant, or that the metric toolkit developed over the past thirty years is wrong. The narrow claim is that the intent reading of those metrics presupposes a line-drawer who is not present in this construction.

Whether deterministic generation is preferable to discretionary redistricting is a political and constitutional question; the algorithm itself cannot answer it.

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